

Comparison of different die-attach techniques

From Fig. 2 it is evident that lead (Pb) based solder is an ideal technology in every way, except for being extremely toxic—making it highly unsuitable. With a good combination of green and yellow indicators, ECA seems to be an allrounder. Although silver (Ag) sintering and TLPS also give excellent performance, they are not practical in terms of cost and workability.

Alternate Solders fail to give the required performance and reliability, even if they excel as an easy successor to traditional lead based solders at low costs. Amidst all the properties of different die-attach techniques against each other, ECA arguably stands out to have the best potential in this market.

Properties of ECAs

Silver is the most preferred material for conductive filler in ECAs owing to its very high conductivity and reliability. But since it has an extremely high cost, other filler materials can be considered where cost plays a huge decisive role.

Fig. 3 shows that silver meets all aspects, apart from density and cost. Copper, on the other hand, simply oxidises too easily, making it unsuitable. Nickel falls widely short on conductivity but makes up with good filler material density.

It can be noted that gold and silver are the only two materials that have the required conductivity and lack in reactivity. However, gold is an

Property	Pb Solder	ECA	Ag Sintering	TLPS	Alternate Solders
Performance					
Reliability					
Die Sizes usable					
Cost					
Workability					
Surface sensitivity					
Moisture sensitivity					
Toxicity					

Fig. 2: Colour-coded comparison chart (Green: favourable, yellow: moderate, red: unfavourable) (Credit: IDTechEx)

even more expensive alternative to silver. With respect to polymer resin materials, most epoxy based ones are used although other varieties are also useful, depending on the exact application and properties required.

Silicone performs better where oscillations need to be accommodated by the adhesive. Polyimides have a high functional temperature but at the cost of more curing temperature, making it expensive. Acrylics also perform well but, unfortunately, do not have a tolerable high functional temperature.

Advantages and disadvantages of ECA

Pros. 1. Low environmental impact due to absence of heavy or toxic metals. This is beneficial in all applications, particularly healthcare or consumer electronics.

2. Ability to connect highly fine pitches. More components can be connected in a small circuit area because the minimum distance required for making a successful connection is low, which is beneficial

for the miniaturisation of electronic components.

3. Mechanically soft due to the large amount of polymer filler that holds everything together. Since ECAs are not structurally based around a brittle metal, they can deform under stress. This characteristic might be beneficial for electronics having fragile components where stress needs to be accommodated during processing, such as with oscillating components, flexible, or in-mould electronics.

4. Lower temperatures are required than solder. Instead of melting, an ECA only needs to be cured at a high temperature for an extended time so that the polymer can react with itself and firm up for holding everything in place. This is beneficial for temperature-sensitive components, like flexible electronics where low-temperature polymer substrates are used.

Cons. 1. Although the temperature required for curing is low, the process is longer than conventional soldering.

2. Silver, a conductive filler used in an ECA, is extremely expensive and a significant drawback for using ECAs on a large scale.

3. Compared to lead-free solder, ECAs require very different equipment to replace lead based technologies and many other similar techniques. Therefore, this may require upfront investment to change all the machines for better compatibility

ICA and ACA

Despite having very similar structures, ICAs tend to have a much higher amount of filler than ACAs. Typically, an ICA might have around 20% polymer matrix and 80% conductive filler to ensure connection is made across the joint. An ACA, on the other hand, might have 20% conductive filler and 80% polymer matrix. Although this allows a small overall connection, it guarantees that no other connections are made at points other than between the connection and the metal pad.

ICA is quite popular, but ACA has a better potential for the future due to benefits such as the ability to find connecting pitches and provide increased flexibility. In many ways, ACA has the same pros and cons as the ICA.